

@relation weather

@attribute outlook {sunny,overcast,rainy}

@attribute temperature {hot,mild,cool}

@attribute humidity {high,normal}

@attribute windy {TRUE,FALSE}

@attribute play {yes,no}

@data

sunny,hot,high,FALSE,no

sunny,hot,high,TRUE,no

overcast,hot,high,FALSE,yes

rainy,mild,high,FALSE,yes

rainy,cool,normal,FALSE,yes

rainy,cool,normal,TRUE,no

overcast,cool,normal,TRUE,yes

sunny,mild,high,FALSE,no

sunny,cool,normal,FALSE,yes

rainy,mild,normal,FALSE,yes

sunny,mild,normal,TRUE,yes

overcast,mild,high,TRUE,yes

overcast,hot,normal,FALSE,yes

rainy,mild,high,TRUE,no

Over////////

BEFORE STARTING ANY EXPERIMENT

STEP 1 — OPEN WEKA

After installing WEKA:

Double click:

WEKA 3.8

You will see:

GUI Chooser

Buttons:

Explorer

Experimenter

KnowledgeFlow

Simple CLI

ALWAYS CLICK:

Explorer

Most experiments happen inside Explorer.

EXPERIMENT 1 — EXPLORE WEKA

DATASET

weather.nominal.arff

FULL STEP-BY-STEP

STEP 1

Open:

WEKA → Explorer

STEP 2

Inside Explorer top-left click:

Open file...

STEP 3

Go where your dataset stored.

Select:

weather.nominal.arff

Click:

Open

STEP 4

Now dataset loaded.

Left side you will see:

Current relation

Below:

Relation name

Instances

Attributes

STEP 5

In center click attribute names one by one:

outlook

temperature

humidity

windy

play

Bottom graph changes automatically.

STEP 6

Now top tabs click one by one:

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

STEP 7

Go to:

Associate

STEP 8

Click button:

Choose

STEP 9

Open:

rules → Apriori

Click:

Apriori

STEP 10

Now click:

Start

Right side output appears.

EXPERIMENT 2 — PREPROCESSING + APRIORI + FP GROWTH

PART A — PREPROCESSING

STEP 1

Open:

WEKA → Explorer

STEP 2

Click:

Open file

Load:

weather.nominal.arff

STEP 3

Stay in:

Preprocess

(Default tab)

STEP 4

Middle section find:

Filter

Click:

Choose

STEP 5

Open folders:

unsupervised

→ attribute

→ Remove

Click:

Remove

STEP 6

Now beside filter click:

Remove

Small box opens.

STEP 7

In:

attributeIndices

type:

1

Means first attribute removed.

STEP 8

Click:

OK

STEP 9

Click big button:

Apply

Now first attribute disappears.

PART B — APRIORI

STEP 10

Top tab click:

Associate

STEP 11

Click:

Choose

STEP 12

Open:

rules → Apriori

Select:

Apriori

STEP 13

Click text:

Apriori

Settings window opens.

STEP 14

Leave default settings.

Click:

OK

STEP 15

Click:

Start

Right side rules appear.

Example:

humidity=normal ==> play=yes

PART C — FP GROWTH

STEP 16

Still inside:

Associate

Click:

Choose

STEP 17

Open:

rules → FPGrowth

Select:

FPGrowth

STEP 18

Click:

Start

Frequent patterns appear.

EXPERIMENT 3 — NAIVE BAYES

DATASET

weather.nominal.arff

FULL GUIDE

STEP 1

Open:

WEKA → Explorer

STEP 2

Click:

Open file

Select:

weather.nominal.arff

STEP 3

Top click:

Classify

STEP 4

Click:

Choose

STEP 5

Open:

bayes → NaiveBayes

Click:

NaiveBayes

STEP 6

Under:

Test options

Tick:

Cross-validation

Set:

10

STEP 7

Click:

Start

STEP 8

Right side output shows:

Accuracy

Correctly classified instances

Confusion matrix

EXPERIMENT 4 — OLAP OPERATIONS

DATASET

weather.nominal.arff

STEP 1

Open WEKA → Explorer

STEP 2

Open:

weather.nominal.arff

STEP 3

Stay in:

Preprocess

SLICE OPERATION

STEP 4

Filter section: Click:

Choose

STEP 5

Open:

unsupervised

→ instance

→ SubsetByExpression

Click:

SubsetByExpression

STEP 6

Click filter name:

SubsetByExpression

STEP 7

In expression type:

ATT1 is 'sunny'

STEP 8

Click:

OK

Then click:

Apply

Now only sunny rows remain.

DICE OPERATION

STEP 9

Again choose:

SubsetByExpression

STEP 10

Expression:

ATT1 is 'sunny' and ATT2 is 'hot'

STEP 11

Click:

Apply

EXPERIMENT 5 — ETL

DATASET

weather.nominal.arff

STEP 1

Open WEKA → Explorer

STEP 2

Load:

weather.nominal.arff

STEP 3

Stay in:

Preprocess

REMOVE ATTRIBUTE

STEP 4

Filter → Choose

Open:

unsupervised

→ attribute

→ Remove

STEP 5

Click:

Remove

Type:

1

Click:

OK

STEP 6

Click:

Apply

NORMALIZE

STEP 7

Again:

Choose

Open:

unsupervised

→ attribute

→ Normalize

STEP 8

Click:

Apply

SAVE FILE

STEP 9

Top right click:

Save

Save transformed dataset.

EXPERIMENT 6 — K-MEANS CLUSTERING

DATASET

iris.arff

STEP 1

Open WEKA → Explorer

STEP 2

Load:

iris.arff

STEP 3

Top click:

Cluster

STEP 4

Click:

Choose

STEP 5

Select:

SimpleKMeans

STEP 6

Click:

SimpleKMeans

STEP 7

Set:

numClusters = 3

Click:

OK

STEP 8

Click:

Start

Clusters generated.

EXPERIMENT 7 — TEXT MINING

DATASET

sentiment.arff

STEP 1

Open WEKA → Explorer

STEP 2

Load:

sentiment.arff

STEP 3

Stay in:

Preprocess

STEP 4

Filter → Choose

Open:

unsupervised

→ attribute

→ StringToWordVector

STEP 5

Select:

StringToWordVector

STEP 6

Click:

Apply

Text converts into numeric words.

STEP 7

Top click:

Classify

STEP 8

Choose:

bayes → NaiveBayes

STEP 9

Click:

Start

Sentiment classification output generated.